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AFFDEX SDK 5.2 Signals

Information about Facial Expression Analysis data/signals provided by Affectiva AFFDEX SDK 5.2.

Affectiva AFFDEX Facial Expression Analysis provides data for Emotions, Facial Expressions, and Behavioral metrics. AFFDEX SDK 5.2 (released in September 2024 and integrated with iMotions 10.1.2+) includes Speech Modeling and associated metrics, along with general improvements.

Note: iMotions will integrate AFFDEX 5.2 by default, which is identical to 5.1 except for the additional metrics (from Speech Modeling). If you are currently using version 4, it is advised that you finish data collection with that SDK version (and avoid updating iMotions). Please contact your Customer Success Manager for more information.

- New Speaking Model provides 'Speech', 'Adaptive Engagement' and 'Adaptive Valence' signals.
- 'Speech' identifies facial movements associated with vocalizations (e.g. chin drop or smile) for improved classification while people are talking
- Adaptive Engagement and Adaptive Valance accounts for speech-related facial movements (standard analysis applies when speech is not detected).
- Other metrics remain available with generrovements and algorithm training
    - AFFDEX SDK 5.1 improved training data, tracking of faces in sub-optimal video conditions, and action unit detection; Added two 'Confusion' and 'Sentimentality' metrics.
    - AFFDEX SDK 4.0

Affectiva AFFDEX Signals

Affectiva AFFDEX provides three types of signals ('Emotion', 'Facial Expression', and 'Behavioral' signals). Each signal type corresponds with a level of processing for facial expression analysis, which includes (1) face detection for Behavioral signals, (2) processing facial landmarks for Facial Expression signals, and (3) processing facial expressions data for Emotion signals. Data is provided for each frame of video where a face is detected.

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You can replay and review your these signals in iMotions Replay windows. Available signals appear in the Signals Menu and will be shown in the Signals Panel (when set to display).



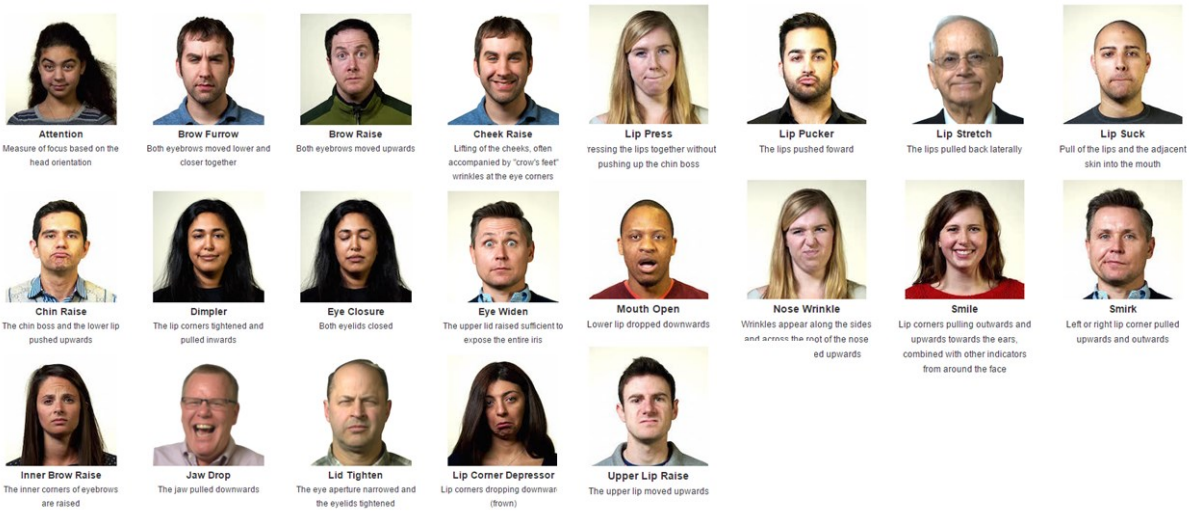
iMotions Replay window with Affectiva AFFDEX signals

- Behavioral - Behavioral metrics based on face detection and facial landmark locations (Pitch, Yaw, Roll, Interoccular Distance)
- Facial Expression - Facial Expression classifications based on detected movement of facial landmarks (Facial Action Units, Attention, Blink Rate; 23 metrics)
- Emotion - Emotion classifications based on combined Facial Expression data (7 Core Emotions and AFFDEX metrics - Engagement, Valence)
- Facial Expression and Emotions signals are scores ranging 0-100, indicating the confidence that a Facial Expression or Emotion was expressed
    - 0=it was not expressed
    - 100=it was expressed
    - 50=50% probability it was expressed

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Facial Expression Signals

Facial Expression signals result from analysis of facial landmarks into 23 channels of data (examples shown below). These signals correspond with Facial Action Unit with additional metrics for Attention and Blink Rate.



Emotion Signals

Emotion signals involve further analysis of Facial Expression signals. These signals indicate Valence, Engagement, Sentimentality, Confusion and each of the 7 core emotions (Joy, Anger, Disgust, Suprise, Fear, Sadness, and Contempt). These signals are calculated from Facial Expression scores that correspond with each emotion (examples shown below)

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Emotion	Increase Likelihood	Decrease Likelihood
Joy	Cheek Raise Smile	/
Anger	Brow Furrow Eye Widen Lid Tighten Upper Lip Raise Chin Raise Lip Press Mouth Open	Inner Brow Raise Outer Brow Raise Nose Wrinkle Smile
Disgust	Nose Wrinkle Upper Lip Raise Chin Raise Mouth Open	Inner Brow Raise Smile
Surprise	Inner Brow Raise Outer Brow Raise Eye Widen Jaw Drop	/
Fear	Inner Brow Raise Outer Brow Raise Brow Furrow Eye Widen Lip Stretcher Mouth Open	Nose Wrinkle Smile
Sadness	Inner Brow Raise Brow Furrow Cheek Raise Lip Corner Depressor Chin Raise	Outer Brow Raise Nose Wrinkle Smile
Contempt	Brow Furrow Smirk	Outer Brow Raise Smile
Sentimentality	Inner Brow Raise Brow Furrow Cheek Raise Lip Corner Depressor Chin Raise Lip Press Lip Suck Smile	Outer Brow Raise Mouth Open
Confusion	Inner Brow Raise Brow Furrow Lid Tighten Nose Wrinkle Dimpler Lip Corner Depressor Lip Pucker Jaw Drop	Smile
Valence	Smile Cheek Raise	Inner Brow Raise Brow furrow Nose wrinkle Lip corner depressor Chin raise Lip press

Valence & Engagement

Valence, Engagement, and other high-level (e.g. Confusion and Sentimentality) are calculated from combinations of facial expression/action unit scores.

Valence consists of the following:

- Inner Brow Raise
- Brow Furrow
- Cheek Raiser
- Nose Wrinkler
- Lip Corner Depressor
- Chin raiser
- Lip Pressor

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- Smile
- Lip Suck & Upper Lip Raise were previously used in SDK 4.0 (removed from SDK 5.2)

Engagement consists of the following:

- Inner Brow Raise (added with SDK 5.2)
- Brow Raise
- Brow Furrow
- Cheek Raise
- Nose Wrinkle
- Lip Corner Depressor
- Chin Raiser
- Lip Press
- Mouth Open
- Lip Suck
- Smile

Sentimentality & Confusion

Sentimentality and Confusion signals were added with AFFDEX 5.1.

Sentimentality is defined as "happy sadness", and is often seen when watching emotionally resonant material. To isolate the AUs that can be associated with sentimentality, Affectiva presented a battery of advertisements that were thought to elicit sentimental responses from viewers, who ranked the videos on their level of perceived sentimentality. The most sentimental ads as defined by self-report were then shown to a larger testing community in the wild while recording their facial expressions over video. Gesture analysis was then conducted to determine what combinations of AUs were most associated with a "Sentimental" reaction. Contrary to other Emotion scores, Sentimentality has several possible combinations of AUs, commonly seen together, that can contribute to a positive score.

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Confusion was created to parse different contexts for the use of Brow Furrow. Brow Furrow can be suggestive of Anger and negative valence in general, but can also suggest mental effort or confusion. AFFDEX's Confusion metric attempts to isolate a particular set of expressions that suggest not only mental workload but also a *lack of comprehension.* While Confusion is not officially part of the EMFACS system, there have been attempts in the scientific literature to qualify the AU combinations associated with Confusion [2]. Here, Affectiva employed a hybrid approach based on what's been published in prior literature but also ran their in-house gesture analysis to define the AUs associated with Confusion. These AUs can include Brow furrow, Inner brow rise, Lid tighten, and Lip Corner Depressor.

Blink Signals

The Blink is detected when the eye closure score gets above and then below a certain threshold during a specific time range. Second, the Blink Rate is calculated on the top of the blink output, and it is the number of blinks happening per minute.

Baselining Options

AFFDEX has historically not required a baseline neutral stimulus because the classifiers employ an internal "causal" or "rolling" baseline. When the rolling baseline is active, each frame being analyzed is also compared to the faces made in past and future frames, to reduce false positives by subtracting prolonged or persistent resting faces.

The assumption behind the rolling baseline is that true, natural facial expressions are dynamic events that manifest and disappear over time, thus persistent expressions are "not natural" and filtered out of the calculation. Internal research has shown that the use of the rolling baseline provides increased accuracy in detecting AUs.

In SDK 4.0, it is not possible to shut off the rolling baseline. With SDK 5.1 (onward), users have the option to toggle the baseline as "on" or "off". When baselining is turned "off", each video frame is evaluated in isolation.

Interpretations

Below is a subset of metrics exported from Sensor Data.

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Expression & Range	Description	Interpretation
Smile (0 - 100)	Lip corners pulling outwards along with other cues from the face (e.g. eyes) combine to indicate a true smile.	A natural, positive smile. Normally a sign of enjoyment.
Brow Furrow (0 - 100)	Both eyebrows move lower and closer together.	Can indicate concentration, confusion or dislike.
Brow Raise (0 - 100)	Both eyebrows move upward.	Spontaneously occurs during surprise.
Lip Corner Depressor (Frown) (0 - 100)	Lip corners drop downward (frown).	Often associated with sadness.
Valence (-100 - 100)	A measure of the positive (or negative) nature of the participant's experience.	-100 negative effect 0 neutral effect 100 positive effect
Engagement (0 - 100)	A measure of emotional responsiveness triggered by the content, whether positive or negative.	Increasing values signify increased emotional engagement.
Attention (0-100)	Measure of point of focus of the user based on the head position	Increasing values signify increased focus.

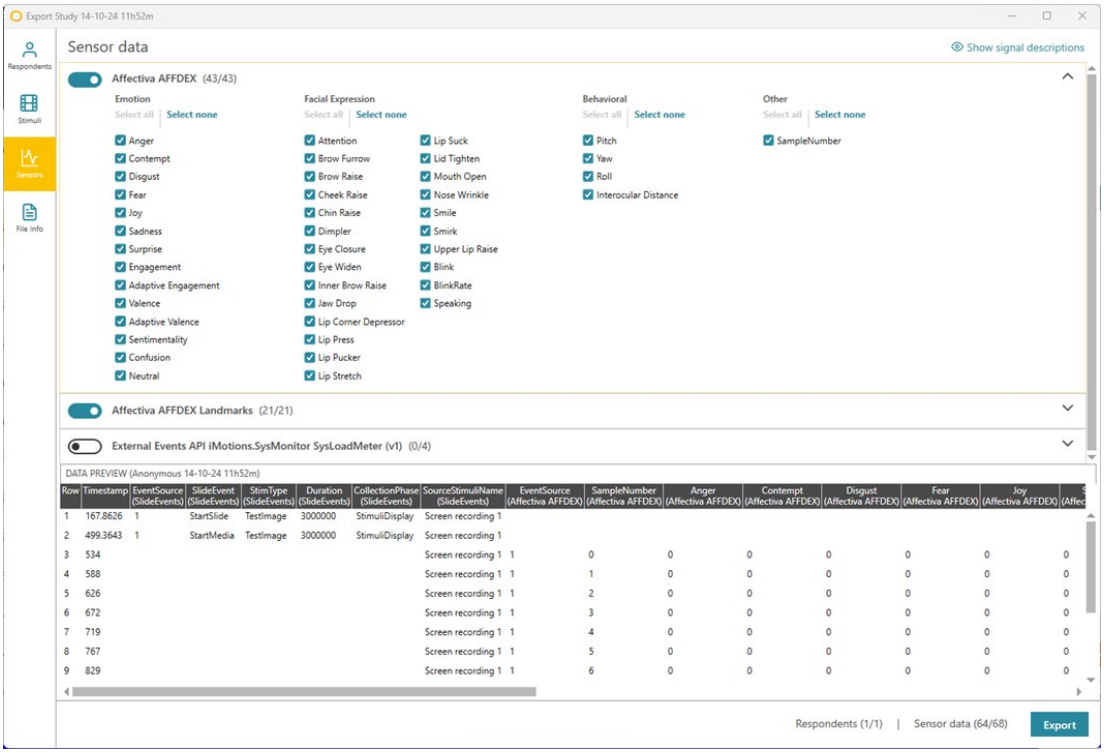
The Emotion scores are still derived from combinations of action unit scores, but these combinations have been overhauled in SDK 5.1 to align more closely with the definitions described in Ekman & Friesen's EMFACS-7. [1] These variables will naturally be improved by the increased performance in the base AU classifiers. In particular, less commonly shown emotions like contempt and sadness are detected with greater confidence and sensitivity.

Exporting Signals

You can use the Sensor Data Export to export your Affectiva AFFDEX signals as .csv files. The Sensor Data Export window can be found by:

1. Right-clicking the study and selecting 'Export'>'Sensor Data'
2. Or by creating an analysis for the study and clicking 'Export' and selecting 'Raw Sensor Data'

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iMotions Sensor Data Export window

AFFDEX Algorithm

Training Data

As of 2022, Affectiva's foundational database over the past twelve years has accumulated a total of 12 million face videos from 90 different countries; a total of over 6 billion facial frames.

Of this primary database, 55,000 videos were then used to build AFFDEX 5.1. Each video contains a unique face for a total of 6.5 million frames. The breakdown of the 55,000 videos is as follows:

- 49,500 videos were used to train the algorithms
- 5,900 videos were used to validate the algorithms
- 8,200 videos were used for testing and determining accuracy.

This subset of the data was specifically chosen to build the SDK to avoid overtraining in the most common action units (smile, brow furrow) and overrepresentation for any one ethnicity or age group. These 55,000 faces were picked to produce the best representation and the greatest diversity of expressions and ethnicities across the entire human race. This has resulted in significant improvements in accuracy for certain demographics, with better recall especially for older African & South Asian populations, and better representation for individuals under 18.

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Similar to SDK 4.0, all ground truth data is analyzed by at least 3 FACS-certified human coders to ensure consistency with the academic standards of the Facial Action Coding System. There must be at least 50% consensus between coders on the presence of a particular action unit or expression before the video can be deemed acceptable for training the SDK.

Previous Version Information

Additional information related to previous versions of the AFFDEX SDK.

Improved Face Trackability & Changes to Landmarks

The face tracker function has been overhauled in 5.1, improving the ability of the SDK to detect the face by 16%. The greatest improvements are for "in the wild collections" that SDK 4 has historically struggled with:

- low light and backlit environments
- videos recorded using a mobile phone
- recordings taken from farther away (at least 40px interoccular distance)
- washed-out faces
- infrared light recordings
- The head angle tolerance has also improved, from 25 degrees to 40.
- Number of facial landmarks used has decreased from 32 to 4 (If your study is heavily reliant on the raw facial landmark coordinate data, it is advised that you keep using SDK 4.0.)

Action Unit Calculation & Improvements in Accuracy

AFFDEX 5.1, as well as 4.0, provides scores for 20 Action Units as defined by the Facial Action Coding System, as well as Attention:

- The AU probabilities were calculated by a combination of support vector machine and recursive convoluted neural networks (CNNs).
- AFFDEX 5.1 has completely rebuilt the 20 AU classifiers from the ground up using CNNs, thus basing the entire SDK on deep learning.
- This has resulted in the majority of AUs showing increased accuracy, particularly for more obscure AUs like Lip Pressor and Cheek Raise. The average ROC-AUC for the new classifiers overall is 91%.

References

[1] Friesen, W. V., & Ekman, P. (1984). EMFACS 7.

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[2] D'Mello, S.K., Craig, S.D., Graesser, A.C. (2009). Multimethod assessment of affective experience and expression during deep learning. *International Journal of Learning Technology*

Affectiva Website: Overview and Technical Details

AFFDEX 2.0: A Real-Time Facial Expression Analysis Toolkit (2022) - external pdf link

AFFDEX SDK: A Cross-Platform Real-Time Multi-Face Expression Recognition Toolkit (2016) - external website link

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